

CHAPTER – XII

APPENDIX

12.1 – Source Code

App.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

@app.route('/about')
def about():
    return render_template('about.html')

@app.route('/dashboard01')
def dashboard01():
    return render_template('dashboard01.html')

@app.route('/dashboard02')
def dashboard02():
    return render_template('dashboard02.html')

@app.route('/dashboard03')
def dashboard03():
    return render_template('dashboard03.html')

@app.route('/story')
def story():
    return render_template('story.html')

if __name__ == "__main__":
    app.run(debug=True)
```



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Base.html

```
<!DOCTYPE html>
<html lang="en">

  <head>

    <meta charset="UTF-8">

    <meta
      name="viewport"
      content="width=device-width, initial-scale=1.0">

    <title>
      Plugging into the Future
    </title>

    <link
      href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.
      min.css"
      rel="stylesheet">

    <link
      rel="stylesheet"
      href="{{ url_for('static', filename='css/style.css') }}">

    </head>

    <body>

      <!-- Navbar -->

      <nav class="navbar navbar-expand-lg navbar-dark shadow">
        <div class="container">

          <a
            class="navbar-brand fw-bold"
            href="/">

            ⚡ Plugging into the Future
```

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```
</a>

<button
  class="navbar-toggler"
  type="button"
  data-bs-toggle="collapse"
  data-bs-target="#menu">

<span class="navbar-toggler-icon">
  </span>

</button>

<div
  class="collapse navbar-collapse"
  id="menu">

<ul class="navbar-nav ms-auto">

  <li class="nav-item">
    <a class="nav-link" href="/">
      Home
    </a>
  </li>

  <li class="nav-item">
    <a class="nav-link" href="/about">
      About
    </a>
  </li>

  <li class="nav-item">
    <a class="nav-link" href="/dashboard01">
      Dashboard 01
    </a>
  </li>

  <li class="nav-item">
    <a class="nav-link" href="/dashboard02">
      Dashboard 02
    </a>
  </li>
```



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```
<li class="nav-item">
  <a class="nav-link" href="/dashboard03">
    Dashboard 03
  </a>
</li>
```

```
<li class="nav-item">
  <a class="nav-link" href="/story">
    Story
  </a>
</li>
```

```
</ul>
```

```
</div>
```

```
</div>
```

```
</nav>
```

```
<!-- Page Content -->
```

```
<main>
```

```
{% block content %}
{% endblock %}
```

```
</main>
```

```
<!-- Footer -->
```

```
<footer class="footer mt-5">
```

```
<div class="container">
```

```
<h5>
```

```
Plugging into the Future
```

```
</h5>
```

BRAVO
— BUILDERS —



Plugging into the Future: An Exploration of Electricity Consumption Patterns

```
<p>

An Exploration of Electricity
Consumption Patterns

</p>

<p>

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</p>
</div>
</footer>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bu
ndle.min.js">
</script>

</body>
</html>
```

Index.html

```
{% extends "base.html" %}
{% block content %}
<section class="hero">
<div class="text-center">

<h1>
Plugging into the Future
</h1>
```

Plugging into the Future: An Exploration of Electricity Consumption Patterns

```
<h3>
An Exploration of Electricity Consumption Patterns
</h3>
```

```
<br>
```

```
<a
href="/dashboard01"
class="btn btn-warning">
```

```
Explore Dashboard
```

```
</a>
```

```
</div>
```

```
</section>
```

```
<div class="container section">
```

```
<div class="row">
```

```
<div class="col-md-4">
```

```
<div class="card">
```

```
<h4>Consumption</h4>
```

```
<p>
```

```
Track electricity usage.
```

```
</p>
```

```
</div>
```

```
</div>
```

```
<div class="col-md-4">
```

```
<div class="card">
```

```
<h4>Visualization</h4>
```

Plugging into the Future: An Exploration of Electricity Consumption Patterns

```
<p>
Interactive analysis.
</p>
```

```
</div>
```

```
</div>
```

```
<div class="col-md-4">
```

```
<div class="card">
```

```
<h4>Forecast</h4>
```

```
<p>
Future energy insights.
</p>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
{% endblock %}
```

About.html

```
{% extends "base.html" %}
```

```
{% block content %}
```

```
<style type="text/css">
```

```
.card{
border-radius:15px;
transition:0.3s;
}
```

```
.card:hover{
transform:translateY(-8px);
}
```

Plugging into the Future: An Exploration of Electricity Consumption Patterns

```
.card img{
  object-fit:cover;
}
</style>
<div class="container section">

  <h1>
    About Project
  </h1>

  <p>

    This project explores electricity
    consumption trends and provides
    visual insights into energy usage.

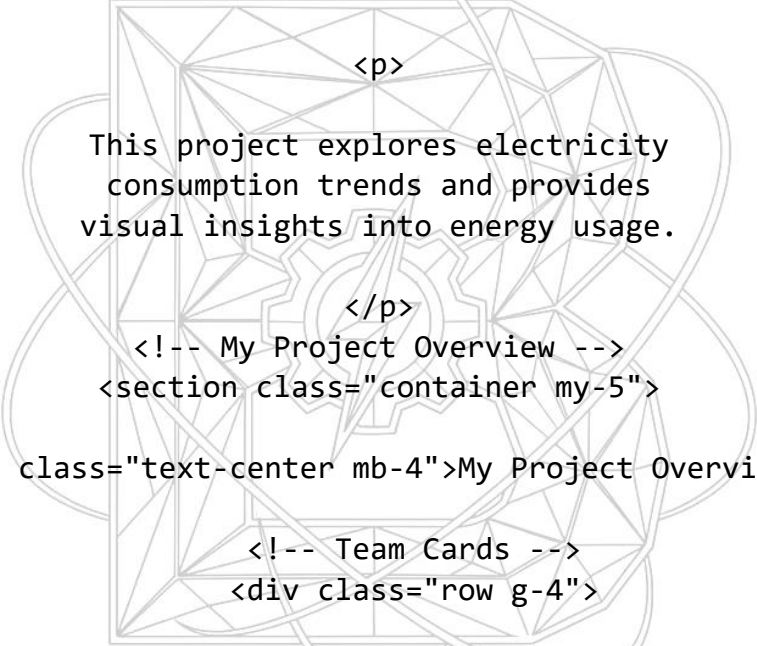
  </p>
  <!-- My Project Overview -->
  <section class="container my-5">

    <h2 class="text-center mb-4">My Project Overview</h2>

    <!-- Team Cards -->
    <div class="row g-4">

      <!-- Team Member 1 -->
      <div class="col-md-4">
        <div class="card shadow h-100 text-center">
          <div class="card-body">
            <h5 class="card-title">VEPPAREDDY GARI LAVA KUMAR
              REDDY</h5>
            <p class="text-muted">Team Leader</p>

            <p class="card-text">
              VEPPAREDDY GARI LAVA KUMAR REDDY is the
              Team Leader of this project.
              He coordinated the project activities, managed team collaboration,
              and guided the development of the Electricity Consumption Analysis
              system to ensure successful project completion.
            </p>
          </div>
        </div>
      </div>
    </div>
  </section>
</div>
```



Plugging into the Future: An Exploration of Electricity Consumption Patterns

```
</p>

<button class="btn btn-primary">
  View Profile
</button>
</div>
</div>
</div>

<!-- Team Member 2 -->
<div class="col-md-4">
<div class="card shadow h-100 text-center">
  <div class="card-body">

    <h5 class="card-title">PAGADALA JAICHANDRA
    DARSHAN</h5>
    <p class="text-muted">Developer</p>

    <p class="card-text">
      PAGADALA JAICHANDRA DARSHAN worked as the
      Developer for this project.
      He was responsible for designing and implementing the web
      application,
      integrating data visualization features, and ensuring smooth
      system
      functionality for users.
    </p>
```

```
<button class="btn btn-success">
  View Profile
</button>
</div>
</div>
</div>
```

```
<!-- Team Member 3 -->
<div class="col-md-4">
<div class="card shadow h-100 text-center">
  <div class="card-body">
```

```
<h5 class="card-title">BELLAMKONDA VENKATA
```

Dashboard.html 1

BELLAMKONDA VENKATA MAHENDRA served as the Data Analyst.

```

    </p>
    <button class="btn btn-warning">
      View Profile
    </button>
  </div>
</div>
</div>
</div>
</section>
</div>
{% endblock %}

```

```
{% block content %}
```

Plugging into the Future: An Exploration of Electricity Consumption Patterns Dashboard 01

State Electricity Analysis

Plugging into the Future: An Exploration of Electricity Consumption Patterns

</h3>

```
<!-- Tableau Container -->
<div class="tableau-container">
```

```
  <div class="tableauPlaceholder"
    id="tableauViz">
```

```
    <object class="tableauViz">
```

```
      <param name="host_url"
```

```
value="https%3A%2F%2Fpublic.tableau.com%2F"/>
```

```
      <param name="embed_code_version"
        value="3"/>
```

```
      <param name="site_root"
        value=""/>
```

```
      <param name="name"
```

```
value="pluggingintotheFutureAnexplorationofelectricityconsumptionpatte
rnsDashboard1/PluggingintotheFutureAnExplorationofElectricityConsumptio
nPatternsDashboard1"/>
```

```
      <param name="tabs"
        value="no"/>
```

```
      <param name="toolbar"
        value="yes"/>
```

```
      <param name="display_static_image"
        value="yes"/>
```

```
      <param name="display_spinner"
        value="yes"/>
```

```
      <param name="display_overlay"
        value="yes"/>
```

```
      <param name="display_count"
```

Plugging into the Future: An Exploration of Electricity Consumption Patterns

```
value="yes"/>

<param name="language"
value="en-US"/>

<param name="filter"
value="publish=yes"/>

</object>

</div>

</div>

</div>

</div>

<style>

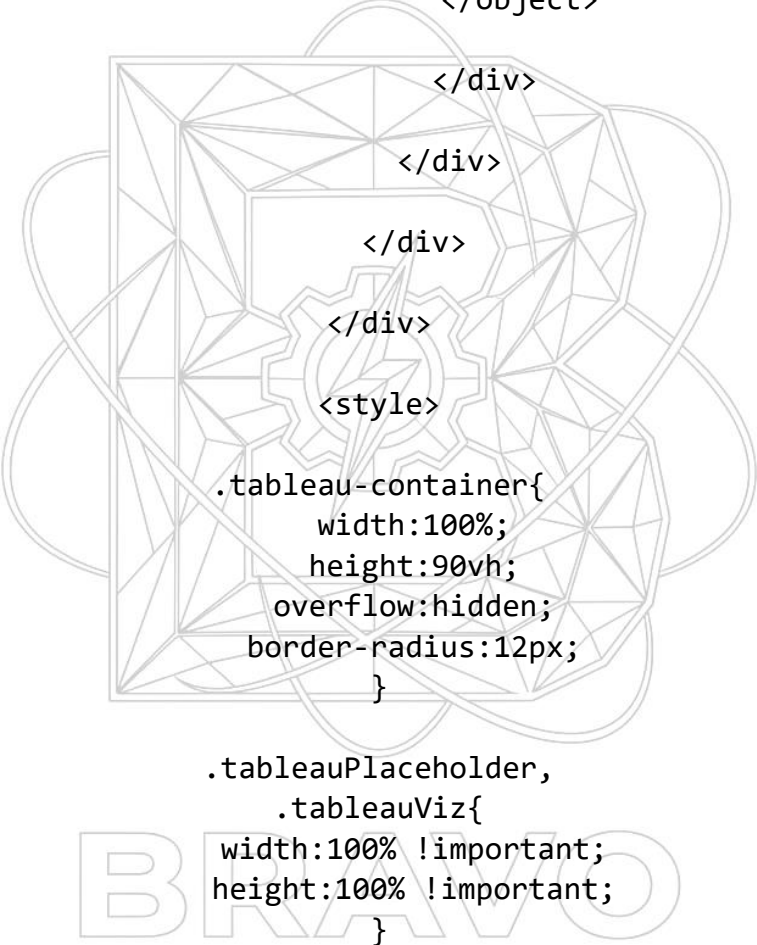
.tableau-container{
width:100%;
height:90vh;
overflow:hidden;
border-radius:12px;
}

.tableauPlaceholder,
.tableauViz{
width:100% !important;
height:100% !important;
}

@media(max-width:768px){
.tableau-container{
height:75vh;
}
}

</style>

<script>
```



Plugging into the Future: An Exploration of Electricity Consumption Patterns

```
var divElement =
document.getElementById("tableauViz");

var vizElement =
divElement.getElementsByTagName("object")[0];

vizElement.style.width = "100%";
vizElement.style.height = "100%";

var script =
document.createElement("script");

script.src =
"https://public.tableau.com/javascripts/api/viz_v1.js";

divElement.appendChild(script);
```

```
</script>
```

```
{% endblock %}
```

Dashboard.html 2

```
...html
{% extends "base.html" %}
```

```
{% block content %}
```

```
<div class="container-fluid section">
```

```
<h1 class="text-center mb-4">
```

Plugging into the Future: An Exploration of Electricity
Consumption Patterns Dashboard 02

```
</h1>
```

```
<div class="dashboard">
```

```
<h3 class="mb-3">
```

Monthly Usage Trends

```
</h3>
```

Plugging into the Future: An Exploration of Electricity Consumption Patterns

```
<!-- Tableau Dashboard -->
<div class="tableau-container">

    <div
      class="tableauPlaceholder"
      id="viz1781510168077">

      <object class="tableauViz">

        <param name="host_url"
          value="https%3A%2F%2Fpublic.tableau.com%2F"/>

        <param name="embed_code_version"
          value="3"/>

        <param name="site_root"
          value=""/>

        <param name="name"
          value="pluggingintothefutureAnexplorationofelectricityconsumptionpatte
rnsDashboard2/PluggingintotheFutureAnExplorationofElectricityConsumptio
nPatternsDashboard2"/>

        <param name="tabs"
          value="no"/>

        <param name="toolbar"
          value="yes"/>

        <param name="display_count"
          value="no"/>

      </object>

    </div>

  </div>

</div>
```



Plugging into the Future: An Exploration of Electricity Consumption Patterns

```
<style>

.tableau-container{
    width:100%;
    height:calc(100vh - 120px);
    overflow:hidden;
}

.tableauPlaceholder,
.tableauViz{
    width:100% !important;
    height:100% !important;
}

.dashboard{
    width:100%;
}

@media (max-width:768px){

    .tableau-container{
        height:80vh;
    }

}

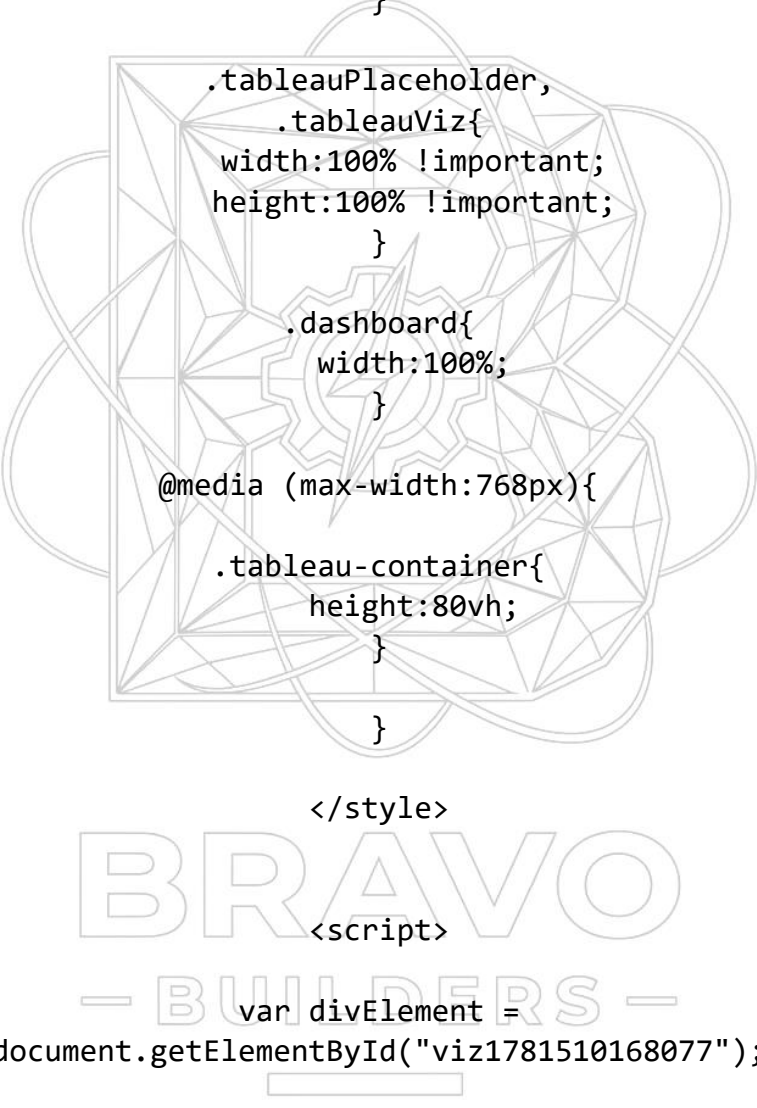
</style>

<script>

    var divElement =
document.getElementById("viz1781510168077");

    var vizElement =
divElement.getElementsByTagName("object")[0];

    vizElement.style.width = "100%";
    vizElement.style.height = "100%";
```



Plugging into the Future: An Exploration of Electricity Consumption Patterns

```
var script =
document.createElement("script");

script.src =
"https://public.tableau.com/javascripts/api/viz_v1.js";

divElement.appendChild(script);

</script>

{% endblock %}
...

Dashboard.html 3
    ``html
    {% extends "base.html" %}
    {% block content %}

    <section class="container-fluid section">

        <div class="container">

            <h1 class="text-center mb-4">
Plugging into the Future: An Exploration of Electricity
Consumption Patterns Dashboard 03
            </h1>

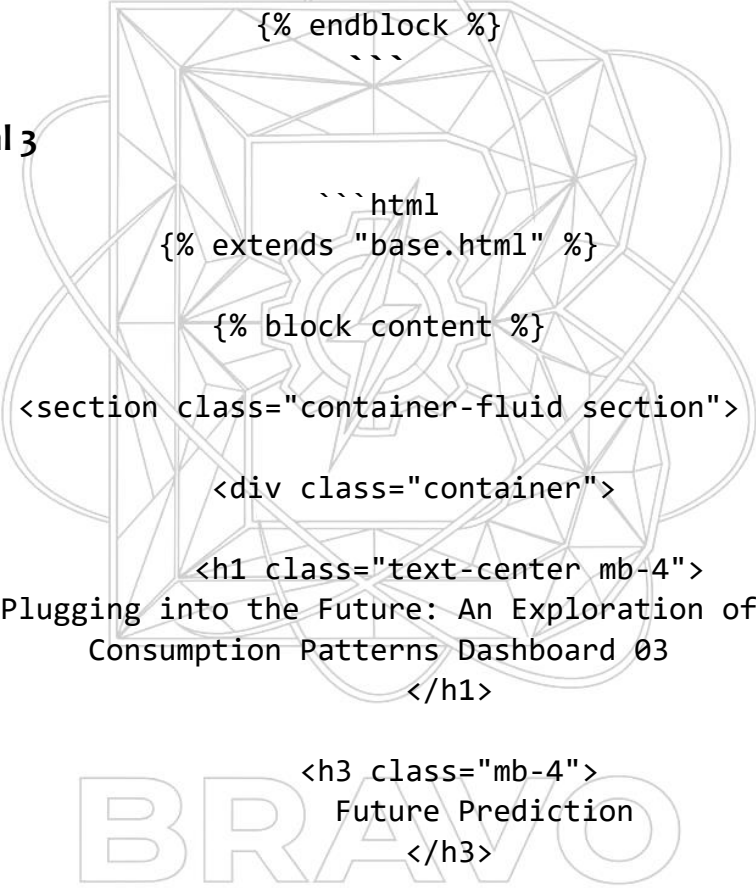
            <h3 class="mb-4">
                Future Prediction
            </h3>

            <div class="tableau-container">

                <div
                    class="tableauPlaceholder"
                    id="tableauViz">

                    <object class="tableauViz">

                        <param
```



Plugging into the Future: An Exploration of Electricity Consumption Patterns

```
name="host_url"
value="https%3A%2F%2Fpublic.tableau.com%2F"/>

    <param
name="embed_code_version"
value="3"/>

    <param
name="site_root"
value=""/>

    <param
name="name"
value="pluggingintotheFutureAnexplorationofelectricityconsumptionpatte
rnsDashboard3/PluggingintotheFutureAnExplorationofElectricityConsumptio
nPatternsDashboard3"/>

    <param
name="tabs"
value="no"/>

    <param
name="toolbar"
value="yes"/>

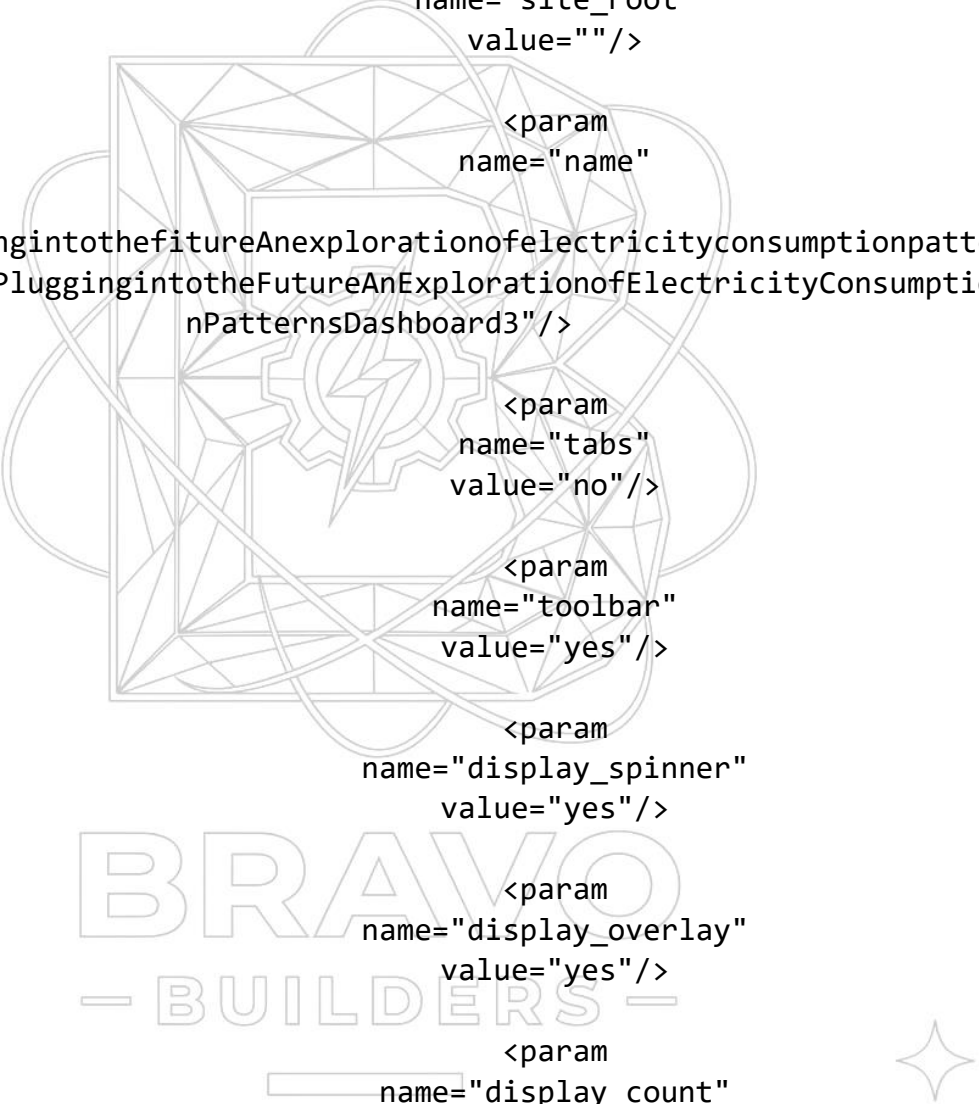
    <param
name="display_spinner"
value="yes"/>

    <param
name="display_overlay"
value="yes"/>

    <param
name="display_count"
value="no"/>

</object>

</div>
```



Plugging into the Future: An Exploration of Electricity Consumption Patterns

```
</div>
```

```
</div>
```

```
</section>
```

```
<style>
```

```
.tableau-container{  
  width:100%;  
  height:calc(100vh - 120px);  
  overflow:hidden;  
}
```

```
.tableauPlaceholder,  
.tableauViz{  
  width:100% !important;  
  height:100% !important;  
}
```

```
@media (max-width:768px){
```

```
.tableau-container{  
  height:80vh;  
}
```

```
}
```

```
</style>
```

```
<script>
```

```
var divElement =  
document.getElementById("tableauViz");
```

```
var vizElement =  
divElement.getElementsByTagName("object")[0];
```

```
vizElement.style.width =  
"100%";
```

Plugging into the Future: An Exploration of Electricity Consumption Patterns

Story.html

```
vizElement.style.height =  
    "100%";  
  
    var script =  
document.createElement("script");  
  
    script.src =  
"https://public.tableau.com/javascripts/api/viz_v1.js";  
  
divElement.appendChild(script);  
    </script>  
    {% endblock %}  
    ...  
    {% extends "base.html" %}  
    {% block content %}  
    <div class="container section">  
        <div class="story">  
            <h1>  
Plugging into the Future: An Exploration of Electricity Consumption  
Patterns Story  
            </h1>  
            <p>  
Electricity powers modern life.  
            </p>  
            <p>  
Through this project,  
we analyze patterns,  
identify trends,  
and visualize how  
energy demand changes.  
            </p>  
        </div>  
    </div>  
    </block %>  
    </endblock %>
```



Plugging into the Future: An Exploration of Electricity Consumption Patterns

```
</p>
<div class='tableauPlaceholder' id='viz1781510540821' style='position:
relative'><noscript><a href='#'><img alt='Plugging into the Future: An
Exploration of Electricity Consumption Patterns '
src='https://public.tableau.com/static/images/pl&#
47;pluggingintothefutureAnexplorationofelectricityconsumptionpatternsS
TORY&#47;Story1&#47;1_rss.png' style='border: none'
/></a></noscript><object class='tableauViz'
style='display:none;'><param name='host_url'
value='https%3A%2F%2Fpublic.tableau.com%2F' /> <param
name='embed_code_version' value='3' /> <param name='site_root'
value='' /><param name='name'
value='pluggingintothefutureAnexplorationofelectricityconsumptionpatte
rnsSTORY&#47;Story1' /><param name='tabs' value='no' /><param
name='toolbar' value='yes' /><param name='static_image'
value='https://public.tableau.com/static/images/pl
&#47;pluggingintothefutureAnexplorationofelectricityconsumptionpattern
sSTORY&#47;Story1&#47;1.png' /> <param name='animate_transition'
value='yes' /><param name='display_static_image' value='yes' /><param
name='display_spinner' value='yes' /><param name='display_overlay'
value='yes' /><param name='display_count' value='yes' /><param
name='language' value='en-US' /><param name='filter'
value='publish=yes' /></object></div>
<script
type='text/javascript'>
var divElement =
document.getElementById('viz1781510540821');
var
vizElement = divElement.getElementsByTagName('object')[0];
vizElement.style.width='916px';vizElement.style.height='591px';
var scriptElement = document.createElement('script');
scriptElement.src =
'https://public.tableau.com/javascripts/api/viz_v1.js';
vizElement.parentNode.insertBefore(scriptElement, vizElement);
</script>
</div>

</div>

{% endblock %}
```

Style.css

```
body{font-family: Arial;background:#f5f7fb}
```

Plugging into the Future: An Exploration of Electricity Consumption Patterns

```
.navbar{background:#002147}

.navbar-brand,.nav-link{color: #fff!important}

.hero{
  height:90vh;
  display: flex;
  align-items: center;
  justify-content: center;
  background: linear-gradient (grab (0,0,0,.6), grab (0,0,0,.6)),
  url("https://images.unsplash.com/photo-1473341304170-971dcc5ac1e")
  center/cover;
  color: #fff
}

. section {padding:70px 0}
. card, Dashboard, story {
  background: #fff;
  border-radius:10px;
  box-shadow:0 5px 20px grab (0,0,0,.1)
}

. card {border: none; padding:20px}
. dashboard {height:500px; padding:30px}
. story {padding:40px; border-radius:15px}

. footer {
  background: #00152e;
  color: #fff;
  padding:20px;
  text-align: center
}
```

